

Distributive energy justice in local electricity markets: Assessing the performance of fairness indicators

Marthe Fogstad Dyngre ^{a,*}, Umit Cali ^{a,b}

^a Department of Electric Energy, Norwegian University of Science and Technology, Trondheim, Norway

^b School of Physics, Engineering and Technology, University of York, York, United Kingdom

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ABSTRACT

Local electricity markets are increasingly seen as a key factor in democratising the energy system. While there has been a significant amount of research on the technical and economic aspects of these markets, it is still not clear how well they align with the social acceptance of potential participants. The concept of fairness in these markets has been gaining attention, but there has not been a clear discussion or definition of what constitutes fairness in a local electricity market context. Furthermore, some fairness indicators have become popular in the literature on local electricity markets, but have yet to be fully analysed on how well they capture fairness in these markets. This study aims to address these issues by formally defining distributive energy justice in the context of local electricity markets and evaluating how well popular fairness indicators perform based on this definition. The indicators are tested on a simulated local electricity market using real consumption data from Norwegian households. The results of this extensive evaluation lead to proposed adjustments to the indicators to make them more suitable for local electricity markets, and a discussion on future research directions for fairness in local electricity markets. When indicators are appropriately aligned with local electricity markets and their definition of fairness, they can trigger changes to market designs that improve distributive energy justice and avoid bias against market participants.

1. Introduction

In recent years, there has been a growing expectation for individual end-users to become more engaged in the energy system, transitioning from mere consumers to active producers of energy [2]. This paradigm shift has given rise to local energy communities and markets, which are viewed as potential avenues for the democratisation of the energy sector. While these markets have been studied from economic and technical perspectives, their alignment with social acceptance criteria remains unclear. The success of a local electricity market depends on all participants understanding and accepting the market's outcomes.

Traditional economic theory, focused on efficiency, often overlooks fairness, assuming it would be achieved through cost-reflectiveness [3]. This viewpoint has also dominated the literature on local electricity markets [4]. However, research in recent decades has recognised the importance of fairness in economic systems [5], acknowledging that even profit-oriented models benefit from considering stakeholders' behavioural characteristics [6]. Ignoring justice in market models may lead to economic instability, social inequity, reduced innovation, and

market distrust. While market models can enhance social well-being, achieving maximum social welfare may still result in unfair distribution among participants [7]. Therefore, it is essential to evaluate local electricity market models not only for their economic and technical feasibility but also for fairness to ensure broad social acceptance.

Energy justice, a framework to examine fairness¹ in the energy sector [8], has gained attention in recent years, focusing on policy-related guidance for energy issues [9]. This research underscores critical factors for the social acceptance of the energy transition, such as fuel poverty or discrimination against vulnerable groups. Analyses related to energy justice often employ a framework based on specific tenets, typically divided into distributive, procedural, and recognition-based justice [10]. *Distributive energy justice* pertains to how the benefits and burdens of the energy transition are allocated among different groups and will be the focus of this article. The most commonly used definition of distributive energy justice is the one proposed by [9]; “*The distribution of benefits and ills on all members of society regardless of income, race etc.*”, where the injustice occurs if “*One or more sections of society are subjected*

* Corresponding author.

E-mail address: marthe.f.dyngre@ntnu.no (M.F. Dyngre).

¹ The terms “justice” and “fairness” are often used interchangeably in this context, and their similarities or differences are a topic of debate [1]. However, these terms can also be seen as mutually reinforcing, as one cannot exist without the other. Therefore, both terms are used judiciously throughout this article.

to an uneven distribution of costs and/or benefits, in terms of geography, demographic etc.” [11]. This perspective is particularly relevant when examining the energy system from a holistic viewpoint but is often criticised for lacking a normative commitment to specific contexts within energy research [12]. This lack of specificity can limit its applicability and effectiveness in addressing issues of energy justice [13].

On the end-user level, most attention within energy justice research has been given to addressing the justice aspects within, between and beyond energy community initiatives [14]. These discussions are often based on an assumption of a spatially bounded community, with clear social and cultural ties and often co-owned energy and storage assets [15]. Many focus on the procedural justice of these initiatives, considering how different members of a community are included in the decision-making process [16]. Recognition energy justice is discussed in terms of the socio-economic barriers that may prevent individuals from participating in energy community initiatives [17], or the issue of gender-related discrimination, particularly in terms of acknowledging active participation [18]. Lastly, distributive energy justice discussions in this context include environmental impacts of energy resource siting [19], local job creation [20] and sustaining a community feeling in rural communities [21]. The distribution of financial benefits is also a prominent discussion within this literature, mainly concerning the distribution of benefits from shared assets [22]. While several of these issues are applicable in the context of a local electricity market as well, there is a lack of explicit discussions of the definitions of energy justice in a local trading setting. This is particularly true for settings that do not necessarily involve a community and where assets are owned individually. Thus, there is still lacking a common understanding of the underlying framework of energy justice for local electricity markets.

An emerging approach to fairness in techno-economic literature on local electricity markets uses quantitative measures to evaluate proposed market structures. Indicators across various fields, including economics, environmental studies, computer science, and telecommunications, have been adapted for use in the context of local electricity markets, serving as measures of equality² and fair resource allocation [24–26]. Predominantly, these indicators are employed after the market clearing to evaluate the performance of market algorithms post-event, assessing the extent of equal treatment among different market participants and identifying potential trading imbalances. However, their ability to capture the heterogeneity and irrational behaviour inherent in local electricity markets is uncertain. No research has been found addressing or comparing the performance of these indicators. Ref. [27] states that the energy justice literature, in general, would benefit from more focus on the use of energy justice indicators and an introduction to the standard use of distributive justice indicators specifically. A comprehensive literature review of fairness in local energy systems [28], not limited to local electricity markets, also concludes with the need of appropriate metrics for evaluation of fairness. In the initial stages of local electricity market design, the use of indicators can facilitate adjustments to models and enable performance comparisons across different models. As fairness indicators mature, fairness should be elevated to a design criterion to promote distributive justice and prevent bias against certain participants. However, to reach this stage, the following must be answered:

- How is fairness addressed in the current literature on local electricity markets?
- How can distributive energy justice be defined in a local electricity market context?
- How do the currently used fairness indicators capture fairness in local electricity markets?

- What fairness indicators should be used in further studies on local electricity markets?

To answer these questions, the article presents a literature review in Section 2, focusing on literature regarding distributive justice interpretations and indicators in local electricity markets. Then, a formal definition of distributive fairness in local electricity markets is proposed in Section 3. Further, the methodology and case study used are presented in Sections 4 and 5 before using the proposed definition in a comprehensive analysis of the performance of the chosen fairness indicators in Section 6. Lastly, policy implications of the redefinition and further use of fairness indicators are discussed in Section 7 before a conclusion and outlook are presented in Section 8.

2. Literature review and contributions

2.1. Interpretations of distributive energy justice in local electricity markets

The principles of distributive energy justice have served as the foundation for numerous studies on local electricity market models, despite the noticeable absence of in-depth discussions regarding the contextual definition. These studies have evaluated market models based on principles of fairness, often referred to as *economic fairness* rather than *distributive justice*. Under economic fairness, more context-specific perceptions such as individual rationality, equality, equity, and coalition stability have been used [29]. The varying interpretations in the literature reviewed are summarised in Table 1.

One of the most common, implicit interpretations of fairness is individual rationality, meaning that the market model itself ensures benefits, although to a varying extent, for all participants [30]. Ref. [31] added transparency and simplicity, in addition to equitable savings and customer acceptance to this definition. The equalising of transparency and fairness was also highlighted in [32]. Ref. [33] linked fairness with each market participant's willingness to pay.

Many argue that a Nash equilibrium solution, associated with game theoretic market models, inherently provides fair outcomes as the individual rationality of the market participants is preserved [39,40]. Ref. [35] quantified fairness by evaluating the distance between the outcome of their proposed market mechanism to that of a Nash outcome. However, while satisfying Nash optimality and efficiency, this solution may deviate significantly from the social optimum [34].

Other measures related to game theoretic models, such as maximum excess and coalition stability, have also been frequently used as fairness measures [29,45]. Solutions that combine individual rationality and equal allocation were considered fair in [30,42]. Both [36,37] defined fairness as equal dissatisfaction with the market outcome for all participants.

The concept of equity, which equates fairness with the allocation of resources based on individual contributions, has also been a prominent approach in the literature on local electricity markets and systems, as reviewed in [28]. These market models are designed to encourage investments in distributed energy resources and flexibility by preferentially allocating resources to their owners. This equity approach was exemplified in the work of [26], who proposed a reputation-based system for matching bids in a peer-to-peer market. They introduced a set of fairness indicators to assess the transactions, with a focus on favouring participants who have earned high reputations. In a similar vein, [29] interpreted fairness within an energy-sharing community as equitable individual fairness. They defined a fair allocation as one that guarantees a non-negative payoff to each community member, calculated based on their individual contribution.

A common method of merit-based cost allocation is the Shapley value, often used in conjunction with cooperative game-theoretic models, and has been used as a guarantee of fairness by [34,41,43]. Ref. [38] utilised a fairness indicator that measures the discrepancy between the cost allocation of a market mechanism and the

² Equity commonly refers to rewarding individuals proportionally to individual contribution. Equality refers to equal rewards irrespective of contribution [23].

Table 1

Summary of interpretations and indicators used in literature on fairness in local energy communities and markets.

Source: Inspired by [29].

Ref.	Interpretation of fairness	Fairness indicator
[24]	Equality	Quality of Service, Quality of Experience, Minimum–maximum
[26]	Equity	Seller Fairness Indicator, Buyer Fairness Indicator, Fairness Factor
[29]	Individual rationality, minimised inequality, coalition stability	Maximum excess, Fairness Index, Quality of Service
[30]	Individual rationality, equality	Participation Willingness Index, Equality Index
[31]	Individual rationality, equity, transparency, simplicity	–
[32]	Equal dissatisfaction, transparency	–
[33]	Willingness to pay	–
[34]	Equity	–
[35]	Individual rationality	Nash social welfare
[36]	Equal dissatisfaction	–
[37]	Equal dissatisfaction, equity	Reputation distribution, Contribution factor
[38]	Equity	Fairness Index
[39]	Individual rationality	–
[40]	Individual rationality, equality	–
[41]	Individual rationality, equity	–
[42]	Individual rationality, coalition stability, equity	–
[43]	Individual rationality, equity	–
[44]	Equity	–
[45]	Incentive compatibility, coalition stability	Maximum excess, stability
[46]	Individual rationality, equality	Quality of Service, Quality of Experience, Minimum–maximum
[47]	Equality	Quality of Service, Quality of Experience, Minimum–maximum
[48]	Equality	Quality of Service, Quality of Experience, Minimum–maximum
[49]	Individual rationality, max–min	Quality of Service
[50]	Equality	Quality of Service, Quality of Experience
[51]	Equality	Quality of Service

Shapley value allocation. However, the Shapley value has scalability issues, possibly making it unsuitable as a fairness measure in real-world implementations of local electricity markets [52]. More scalable approximations have been suggested, for example, by [44]. Ref. [45] compared the Shapley value with other game-theoretical allocation mechanisms, demonstrating that the Shapley value does not necessarily ensure a fair allocation.

The most explicit approach to fairness in the local electricity market literature is through the use of indicators [29]. These indicators can be inspired by other research fields or specifically created for the occasion. Among them, three have gained momentum in recent studies, as demonstrated by examples such as [24,46–51]. These are the Quality of Experience (QoE), Quality of Service (QoS) and Minimum-Maximum (MinMax) fairness indicators, commonly used within telecommunication research. These indicators are mostly used based on an equality-based perception of fairness [28]. Additionally, the Gini coefficient, widely used in economics to represent income or welfare inequality, has been adapted to a local electricity market context in the form of the Equality Index (EI) [30]. The EI measures how income distribution between peers in a market deviates from a perfectly equal distribution. However, despite these indicators' increasing popularity, their relevance to the local electricity market setting has never been explicitly discussed. In fact, some studies have acknowledged that the indicators may not accurately reflect what is considered fair in a local electricity market context but have continued using them in lack of more suitable alternatives [24].

2.2. Contributions

Despite numerous approaches to economic fairness, the reviewed literature reveals a notable absence of explicit discussions on distributive energy justice within the context of local electricity markets. While several fairness indicators have been recognised in the literature, their actual performance and relevance have not been thoroughly examined.

A recent literature review on fairness in local energy systems [28] concludes with the need for both developing new standards for defining fairness and appropriate evaluation metrics in a local energy setting. Given the findings in this literature review, it is clear that before continuing the development of new indicators, a step back to address the underlying definitions and evaluation of existing metrics adequately is needed. This study seeks to bridge these gaps by conducting a comprehensive evaluation of existing fairness indicators and exploring the concept of distributive energy justice in a local electricity market setting. Four of the most common indicators are addressed: QoE, QoS, MinMax and EI. Other indicators referenced in Table 1 are not considered in this study due to their specificity to game-theoretic models or their lack of generalisability beyond reputation-based market structures. It is important to note that this article does not question the mathematical correctness of the indicators but rather the relevance of use in its current form in a local electricity market context. Additionally, this article exclusively focuses on distributive fairness, specifically the fair allocation of costs and resources. The contributions of the study are as follows:

- A proposal of a formal definition of distributive energy justice in a local electricity market context.
- An in-depth evaluation of the recognised fairness indicators MinMax, QoS, QoE and EI, focusing on their effectiveness in representing distributive fairness across nine different case studies within a local electricity market.
- A proposal of adjustments to the MinMax, QoS and EI indicators to better fit the proposed definition of distributive justice.

3. Defining distributive energy justice for local electricity markets

The concept of fairness within a local electricity market is multifaceted, with cultural, social, and geographical factors playing significant roles. Despite this complexity, it is possible to delineate certain

universal principles. In the context of this study, a local electricity market is envisioned as a trading platform that extends beyond the confines of a local energy community. The envisioned market platform incorporates mechanisms that allow market participants to freely enter or exit based on their perceptions of market fairness. This implies that participants who deem the market conditions as unfair have the liberty to discontinue their involvement. Fairness can be operationalised through both subjective perceptions of market conditions and objective fairness indicators. Participants can assess fairness based on their personal experiences and observations within the market. Additionally, fairness indicators can be effectively communicated through visual representations in a local market app or platform. To aid potential new participants in their decision-making process, the platform can present historical statistics of fairness indicators, along with a clear explanation of the agreed-upon definition of distributive energy justice. This approach ensures that all participants have access to transparent and comprehensible information, enabling them to make informed decisions about entering or exiting the market. Fairness viewpoints derived from the social connections present in community structures are ignored in this study, as well as the perspectives of the market or system operators.

Past research has largely focused on local electricity market scenarios involving only prosumers or prioritised prosumer profitability as the primary objective of market-clearing mechanisms. On the other side, most energy justice literature concerning end-users focuses on marginalised consumer groups. However, for the sustainable implementation of such trading platforms in real-world settings, it is imperative that participation yields benefits for both prosumers and consumers. This approach ensures that the energy transition does not become exclusive to those who can afford to invest in distributed energy resources, thereby also preventing it from becoming a financial burden for those who cannot afford such investments through subsidising feed-in tariffs or increased grid costs.

With these considerations serving as the foundational motivation and objective for the establishment of local electricity markets, and to compensate for the lacking definitions in the existing literature, we propose the following formal definition of distributive energy justice in this context:

Definition. Distributive energy justice in a local electricity market can be formally defined as the state in which all market participants benefit from participating in the local market compared to other alternatives. This state is achieved when prosumers and flexibility providers receive rewards that are proportional to their respective contributions to the energy balance within the local electricity market. Simultaneously, it ensures that consumers benefit from an equal distribution of resources. This dual focus on proportional rewards and equal allocation underscores the balance between incentivising investment and promoting accessibility within the market.

This state can be achieved through allocating equal *shares* of available surplus traded within the market to all the prosumers and battery owners and equal *volumes* of covered demand to the consumers. By shares, we mean the proportion of net surplus that is sold within the local electricity market instead of to the retailer. From a prosumer perspective, one may argue that an equal contribution to the local energy balance should yield an equal share of market trades to ensure viability and promote further investments in distributed energy resources. Rewarding the larger prosumer with an equal share to the smaller ones, instead of equal volume or equal monetary benefit, can achieve this. However, for consumers, this approach could potentially foster undesirable behaviours such as increased consumption, leading to ethical dilemmas, particularly given the correlation between energy consumption and socioeconomic status. Thus, allocating equal volumes of energy to the consumers, regardless of their consumption, would be considered more fair.

Note, however, that we consider this definition regardless of the market clearing mechanism used. Allocation mechanisms enforcing equal volume or equal share distribution can be implemented, thus achieving perfect fairness from a consumer perspective according to the definition proposed here. However, such mechanisms may interfere with other objectives of the local electricity market or with the fairness perspectives of other market participants. This definition should thus be perceived as a guideline for designing market mechanisms rather than a strict rule.

A local electricity market's ability to conform to this definition of distributional energy justice can be continuously assessed using the fairness indicators, whose compatibility will be addressed further in this study.

4. Methodology

The overall concept envisioned in this article is illustrated in Fig. 1. The fairness indicators, determined by the definition of distributive energy justice within local electricity markets, evaluate the outcome of the market and pricing mechanisms. The outcome of the indicator assessment is provided to a policymaker who makes the necessary adjustments to the local electricity market to align with the fairness criteria. This can include more extensive regulatory adjustments made by governmental agencies or local adjustments made by the local market operator to adhere to local conditions if the fairness indicators over time imply, e.g. discrimination against groups of market participants. This last iterative step is not included in the methodology of this paper but is rather an envisioned practical implication of mature fairness indicators. However, some possible policy implications are discussed at the end of the paper.

To properly assess the fairness indicators' behaviour, they are all tested on the same local electricity market model and case studies. The market is centrally cleared using a linear program, and local trading prices are determined using the supply-demand ratio pricing mechanism. Both the market and pricing methods were chosen for their simplicity and transparency, proven by the authors in a comparative study with more complex pricing mechanisms in previous work [47]. Unlike the previous study, which provided an in-depth analysis of different pricing mechanisms, this study focuses on the performance of fairness indicators. Thus, although using the same simulative framework, the market model is not a contribution of this study in itself but rather a tool to analyse the fairness indicators. As one of the main objectives of this study is to evaluate the fairness indicators' abilities, transparent methods allow linking their behaviour more easily to the market clearing events. Furthermore, the central market clearing approach exemplifies a frequent goal of local electricity market models in the literature, which is to maximise social welfare. Note that the demonstrated clearing approach is a pure transactional market model and does not consider any grid impacts. Also, note that the intention of this study is not to evaluate the fairness of the analysed market itself but rather the indicators' ability to capture potentially unfair situations within the market. This will be based on the definition provided in Section 3.

4.1. Market model

In this subsection, the equations related to the market and pricing models used in this study are presented.

4.1.1. Central market clearing method

The market is cleared with the objective of minimising the total cost of the local electricity market as a whole. All variables in the model are non-negative. As defined in (1), the objective function includes minimising the net cost of importing g_{ht}^{imp} from the wholesale market at the cost of the spot price C_t^{spot} and other levies C^g , and exporting g_{ht}^{exp} to the retailer for the feed-in tariff C^{FIT} . The feed-in tariff is chosen here

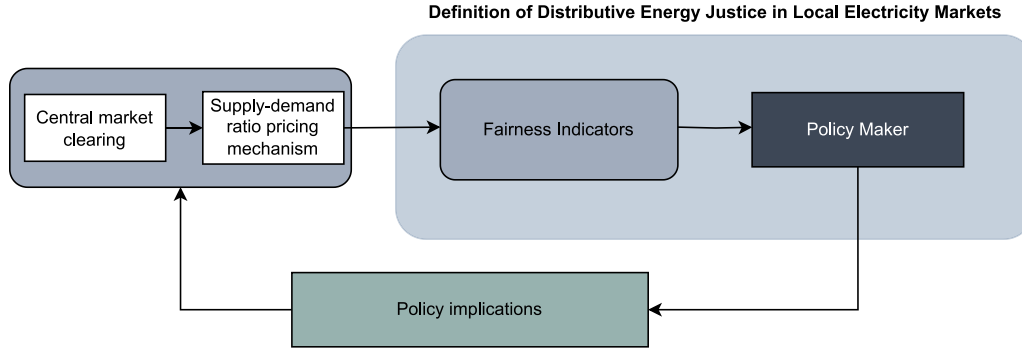


Fig. 1. Flowchart of the envisioned purpose of fairness indicators.

as a representation of the price a prosumer would get if selling their surplus electricity outside of the local electricity market. The variable can, however, be replaced by any other lower pricing bound, as long as it is lower than the consumer buying price. This is true for all further use of the variable in this paper. As the cost of buying and selling within the local electricity market is assumed to be equal, the net costs of these internal trades are assumed to be zero and are thus not included in the objective function.

$$\min \sum_h \sum_t \left((C_t^{spot} + C^g) \cdot g_{ht}^{imp} - C^{FIT} \cdot g_{ht}^{exp} \right) \quad (1)$$

The internal trades are, however, assumed to be balanced within the local electricity market, and the amount bought by house h from peer p , i_{hpt} , is restricted to equal the amount sold by peer p to house h , x_{pht} in time step t , by (2).

$$i_{hpt} = x_{pht}, \quad \forall p \neq h, \forall ht \quad (2)$$

The sum of these transactions should also equal the total amount traded by each market participant. In (3) and (4), i_{ht}^{LM} and x_{ht}^{LM} represent the total bought and sold, respectively, within the local electricity market by house h in time step t .

$$i_{ht}^{LM} = \sum_{p \neq h} i_{hpt} \quad \forall ht \quad (3)$$

$$x_{ht}^{LM} = \sum_{p \neq h} x_{pht} \quad \forall ht \quad (4)$$

If a house owns a battery, additional constraints are required to control its operation. The state-of-charge (SoC) of each battery, represented by the variable b_{ht}^{soc} , is dependent on the SoC of the previous timestep (b_{ht-1}^{soc}) and the amount charged, b_{ht}^{ch} , and discharged, b_{ht}^{dis} , in the current time step, as defined in (5). The charging and discharging efficiencies are represented by η^C and η^D .

$$b_{ht}^{soc} = b_{ht-1}^{soc} + \eta^C \cdot b_{ht}^{ch} - \frac{1}{\eta^D} \cdot b_{ht}^{dis} \quad \forall ht \quad (5)$$

The minimum ($\underline{soc}$) and maximum ($\overline{soc}$) SoC level of the batteries are set by (6). Likewise, the maximum electricity charged (α) and discharged (β) from the battery per time step are defined by (7).

$$\underline{soc} < b_{ht}^{soc} < \overline{soc} \quad \forall ht \quad (6)$$

$$0 < b_{ht}^{dis} < \beta; 0 < b_{ht}^{ch} < \alpha \quad \forall ht \quad (7)$$

Finally, each house is constrained by the energy balance of (8), ensuring that the amount of energy consumed or exported equals the amount of energy produced or imported. PV_{ht} represents the amount of energy produced by the photovoltaic (PV) panels at house h in time step t , while D_{ht} represents the demand.

$$g_{ht}^{imp} + i_{ht}^{LM} + PV_{ht} + b_{ht}^{dis} = D_{ht} + x_{ht}^{LM} + g_{ht}^{exp} + b_{ht}^{ch} \quad \forall ht \quad (8)$$

4.1.2. Supply-demand ratio pricing mechanism

In the supply-demand ratio pricing mechanism, the internal local electricity market trading prices are set ex-post based on the ratio between the surplus available and the net demand in the local electricity market. This ratio, s_t , is defined in (9).

$$s_t = \frac{\sum_h (g_{ht}^{exp} + x_{ht}^{LM})}{\sum_h (g_{ht}^{imp} + i_{ht}^{LM})} \quad \forall t \quad (9)$$

The ratio is used to set the local electricity market trading price between an upper and lower price bound. It is assumed that in order to maintain a stable market situation, the local price should not exceed what consumers otherwise would get from the retailer, i.e. the retailer's selling price. Likewise, no prosumer would sell their excess electricity for a lower price than what they could get elsewhere. Hence, the lower bound is set to equal the feed-in tariff. The local electricity market price, λ_t , is calculated as defined in (10).

$$\lambda_t = s_t \cdot C^{FIT} + (1 - s_t) \cdot (C_t^{spot} + C^g) \quad \forall t \quad (10)$$

If $s_t \geq 1$, the local trading price is equal to the feed-in tariff.

4.2. Analysed fairness indicators

In this subsection, the analysed fairness indicators, along with their proposed alterations, are presented. MinMax, QoS, QoE and EI are first presented with the definitions most commonly used in previous local electricity market literature. The first two, MinMax and QoS, are here used in relation to the energy allocation within the market, while QoE and EI are used to assess the economic distribution between the market participants.

4.2.1. Original minimum maximum indicator

The MinMax indicator measures the difference between the minimum and maximum imports from the wholesale market, as defined in (11).

$$\text{MinMax}_t = \frac{\min_h (g_{ht}^{imp})}{\max_h (g_{ht}^{imp})} \quad \forall t \quad (11)$$

With a MinMax value of one, all participants rely equally on the system outside of the local electricity market.

4.2.2. Original quality of service

QoS assesses allocation fairness using Jain's index [53] and is here defined by (12), where H is the total number of market participants.

$$\text{QoS}_t = \frac{(\sum_h (x_{ht}^{LM} + i_{ht}^{LM}))^2}{H \cdot \sum_h (x_{ht}^{LM} + i_{ht}^{LM})} \quad \forall t \quad (12)$$

A QoS value of one indicates an equal allocation of locally traded electricity among all market participants.

4.2.3. Original quality of experience

The QoE indicator measures the differences in observed electricity prices within the local electricity market. This observed price, q_{ht} , is defined by (13) and is derived using the price at which participants can cover their net demand.

$$q_{ht} = \frac{g_{ht}^{imp} \cdot (C_t^{spot} + C^g) - g_{ht}^{exp} \cdot C^{FIT} + \lambda_t \cdot (i_{ht}^{LM} - x_{ht}^{LM})}{D_{ht} - PV_{ht} + b_{ht}^{ch} - b_{ht}^{dis}} \quad \forall ht \quad (13)$$

The QoE indicator is then defined in (14) as the ratio between the standard deviation of the perceived prices, σ , and the maximum deviation, $\sigma_t^{max} = (C_t^{spot} + C^g) - C^{FIT}$. As $(C_t^{spot} + C^g)$ and C^{FIT} represent the upper and lower bound of the local trading price, respectively, the difference between these prices reflects the maximum possible deviation of perceived prices among the market participants.

$$QoE_t = 1 - \frac{\sigma_t}{\sigma_t^{max}} \quad \forall t \quad (14)$$

When all participants experience equal perceived prices, the QoE gets a value of one.

4.2.4. Original equality index

The EI is defined in (15), where y_h represents the cost saving of participant h for participating in the local electricity market relative to only participating in the wholesale market.

$$EI = 1 - \left(\frac{2 \sum_{h=1}^H h y_h}{H \sum_{h=1}^H y_h} - \frac{H+1}{H} \right) \quad (15)$$

The cost saving is sorted in ascending order, meaning that index h is reordered such that $h = 1$ equals the house with the lowest savings and $h = H$ equals the house with the highest savings. In contrast to the conventional Gini index, the index has a reversed definition, meaning that a value nearer to one denotes a more equal distribution of cost savings. This is done to correlate with the definitions of the other indexes in this paper and, thus, to ease the readability of the results. This is also in line with the definition used in [30].

4.2.5. Alternative minimum maximum indicator

A MinMax value of one would indicate that all market participants need to import the same amount of energy from the wholesale market. In cases with high seasonal variations in load and PV generation, one could argue that the current definition of this indicator has limited value. The vast electricity consumption usually occurs during winter due to electric heating or in the afternoon, when there is limited activity in a PV-only local electricity market. Thus, whatever resource allocation within the local electricity market would have a limited effect on the total import of each market participant.

However, simple redefinitions can still make it relevant, e.g. switching the focus from grid import to the energy trading within the local electricity market. Following the definition in Section 3, an equal volumetric distribution among consumers and a proportional distribution among prosumers could promote a sense of fairness among market participants. In this context, it makes more sense to have a separate indicator for different market participant categories rather than a joint one, as this is a direct proportional indicator. For example, an indicator value of 0.4 means that the least rewarded participant receives 40% of what the most rewarded participant gets. This is different from an interpretation that 40% of the market perceives the market as fair, as is the case with other indicators. An alternative MinMax for prosumers is proposed in (16), whereas (17) proposes one for consumers.

$$MinMax^{pro} = \frac{\min \left(\sum_t \frac{x_{ht}^{LM}}{x_{ht}^{LM} + g_{ht}^{exp}} \right)}{\max \left(\sum_t \frac{x_{ht}^{LM}}{x_{ht}^{LM} + g_{ht}^{exp}} \right)} \quad \forall h \in P \quad (16)$$

$$MinMax^{con} = \frac{\min \sum_t i_{ht}^{LM}}{\max \sum_t i_{ht}^{LM}} \quad \forall h \in C \quad (17)$$

The prosumer MinMax measures the ratio between the prosumer selling the smallest share of its surplus within the local electricity market and the prosumer selling the largest share. The consumer MinMax measures the ratio between the smallest volume bought from the local electricity market and the largest.

4.2.6. Alternative quality of service

The current QoS indicator defines equal volume traded by all participants as the most fair, an approach that has been questioned by its users but not thoroughly addressed [24]. Referring to the definition of distributional fairness in local electricity markets proposed in Section 3 and the following discussion, an alternative version of QoS might be more appropriate. The original QoS indicator may still be relevant when analysing the consumers separately, but it should consider the equality of shares distributed among the prosumers rather than volumes. Separate indicators for the participant groups could be used. However, they should still function as a metric for overall social fairness. Thus, a joint indicator is proposed in (18).

$$QoS_t^{new} = \rho \cdot \frac{\left(\sum_h^P \left(\frac{x_{ht}^{LM}}{x_{ht}^{LM} + g_{ht}^{exp}} \right) \right)^2}{P \cdot \sum_h^P \left(\frac{x_{ht}^{LM}}{x_{ht}^{LM} + g_{ht}^{exp}} \right)^2} + \mu \cdot \frac{\left(\sum_h^C i_{ht}^{LM} \right)^2}{C \cdot \sum_h^C (i_{ht}^{LM})^2} \quad \forall t \quad (18)$$

Here, P is the number of prosumers in the local electricity market and C is the number of consumers. ρ is the ratio of prosumers in the market (P/H) and μ is the ratio of consumers (C/H).

4.2.7. Alternative equality index

In line with the proposed definition in Section 3, one should also differentiate between the prosumers and consumers when considering the issue of equitable distribution of cost savings. The current definition of the EI arranges cost savings based on monetary values. However, this approach may not necessarily result in fairness when applied to prosumers who have different investment costs and capacities. Thus, an equal cost saving increase in percentage compared to not participating in the market for the prosumers and an equal monetary cost saving for consumers are relevant for this indicator as well. The alternative formulation is proposed in (19).

$$EI^{new} = \rho \cdot \left(1 - \left(\frac{2 \sum_{h=1}^P h \left(\frac{y_h}{y_{noLEM}} \right)}{P \sum_{h=1}^P \left(\frac{y_h}{y_{noLEM}} \right)} - \frac{P+1}{P} \right) \right) + \mu \cdot \left(1 - \left(\frac{2 \sum_{h=1}^C h y_h}{C \sum_{h=1}^C y_h} - \frac{C+1}{C} \right) \right) \quad (19)$$

The first part, multiplied by the prosumer ratio in the local electricity market ρ , finds the distribution of relative cost savings among the prosumers. The relative cost saving is defined as the monetary cost saving y_h divided by the costs of not participating in the local electricity market, y_{noLEM} . The EI for the consumers, multiplied by the consumer ratio μ , is calculated in the same manner as the original EI.

5. Case study

In this study, a simulation of the market has been conducted using nine different combinations of PV and battery numbers. Table 2 summarises the combinations in a matrix that shows which house owns which asset in each combination. This comprehensive set of case studies is chosen to have a solid foundation when drawing conclusions about the indicators' performance. In all cases, prosumers are equipped with PV capacities of either 5 or 10 kWp in accordance with their

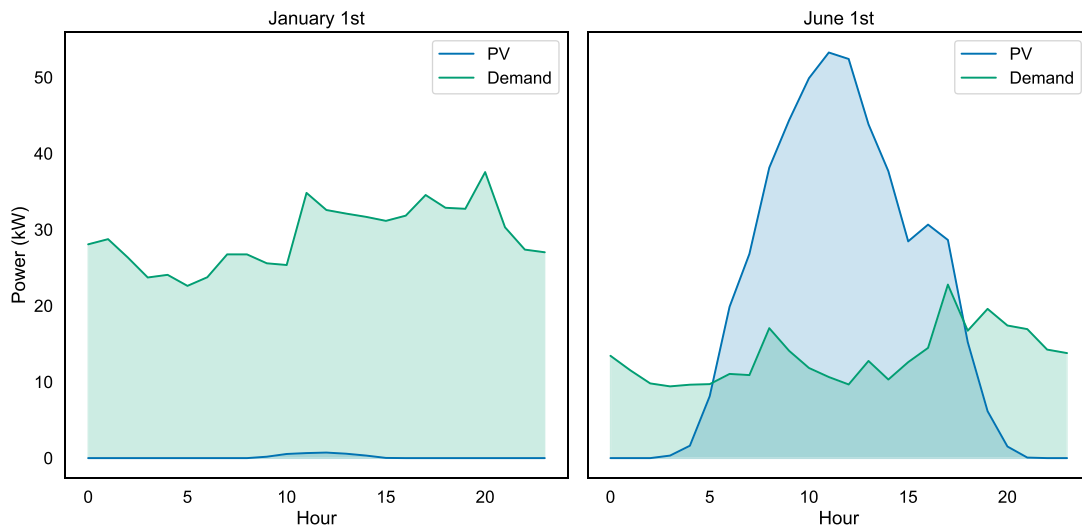


Fig. 2. Aggregated PV and demand profiles of the local electricity market for one random day during the winter and one in the summer.

Table 2

Allocation of PV panels and batteries in the case studies.

		Number of batteries					
		0	1		5		
Number of PV owners	10	House 1–10	House 1–10	House 1	House 1–10	House 1–5	
	5	House 1–5	House 1–5	House 1	House 1–5	House 1–5	
	1	House 1	House 1	House 1	House 1	House 1–5	

Table 3

Annual demand and installed PV capacity of each house.

	House nr.									
	1	2	3	4	5	6	7	8	9	10
Demand [MWh]	29	32	14	13	17	20	27	18	5	9
PV capacity [kWp]	10	10	10	5	5	5	10	10	5	5

peak load requirements, as summarised in Table 3. All battery systems exhibit uniform 6 kWh/3 kW storage capacities, with a 90% round-trip efficiency.

The data is based on real-life historical data obtained from Norwegian households between October 2020 and March 2022 [54]. Only ten households from the Oslo area were chosen to keep an overview of the individual effects of the local electricity market in the fairness evaluation.

None of the households had PVs installed at the time of the data collection, and this is added to the case study. PV generation data for the area were obtained using the PVGIS online tool [55], using irradiation data for 2020. The aggregated PV profiles are plotted against the aggregated demand profiles of the local electricity market in Fig. 2. One arbitrary winter and summer day is selected to highlight the significant seasonal fluctuations in the local energy balance. The day-ahead electricity spot prices from 2021 for the NordPool price area NO1 were retrieved from the same data set as the consumption data. The grid tariff was set to 4.48 ¢/kWh, aligning with the variable term determined by the local distribution system operator for the year 2021 [56]. In compliance with current Norwegian regulations, prosumers are obliged to enter into a contractual agreement with a chosen retailer for the sale of excess electricity to the grid. Predominantly, these contracts feature a feed-in tariff equivalent to the day-ahead spot price, which is the parameter used in this case study.

6. Assessing the performance of fairness indicators

In this section, the four indicators' ability to capture different situations in the given local electricity market is thoroughly analysed. The evaluation is done by analysing the outcome of the market clearing and comparing how the indicators behave in accordance with the market situation. The evaluation criteria are primarily aligned with the proposed definition of distributive energy justice. The two indicators related to the allocation of traded energy within the local electricity market are addressed first, followed by the two indicators related to economic distribution.

6.1. Energy allocation

Investigating the hourly trades of the local electricity market in detail, it is observed that the central market clearing mechanism allocates the trades to only a few consumers each time step. As the objective is to minimise the aggregated total costs of the market participants, it allocates this to the consumers with the highest consumption. However, over a longer period, the allocation evens more out among the consumers, as is represented by the monthly averages in Fig. 3. One can assume that the billing of the local electricity market will be monthly based, like the current retailer practice, at least in early-stage implementation, without any real-time trading. Thus, a fairness evaluation of the monthly allocation should be sufficient, and the two following indicators are calculated as monthly averages.

The average monthly electricity traded within the local electricity market throughout the year is summarised in Fig. 3. Here, the bars represent the net volume sold (positive) or bought (negative) by each house. The dots represent the net shares traded, meaning the share of available surplus sold within the local electricity market for the prosumers (positive) and the share of demand covered by local trade for the consumers (negative). As can be observed, the shares and volumes give quite different pictures of the allocation in some cases. For example, in the cases with five prosumers, House 9 gets

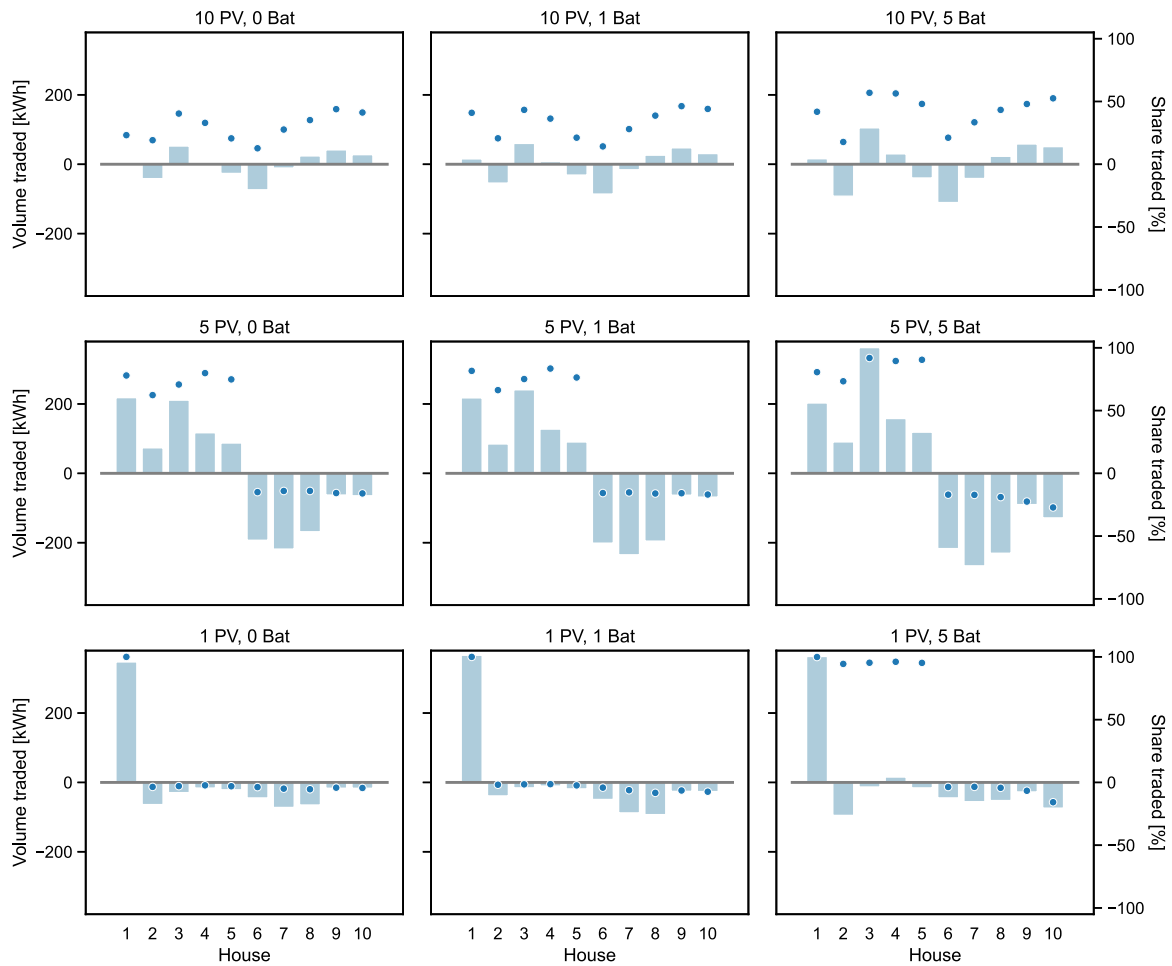


Fig. 3. Shares (dots) and volumes (bars) of each participant's average monthly transactions in the local electricity market. Positive values represent sold volumes or share of surplus sold, negative values represent bought volume or share of demand covered. PVs and batteries are allocated to the lowest house number available for each increment.

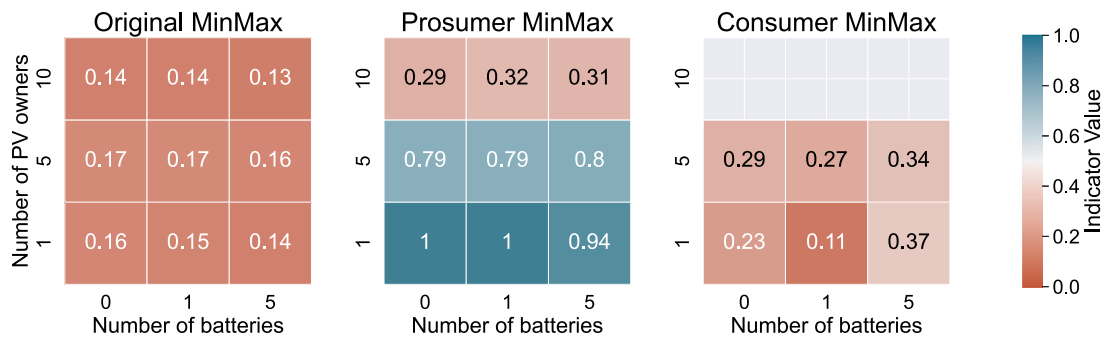


Fig. 4. Monthly average original and separate MinMax values for prosumers and consumers. A value of 1 indicates a completely fair market.

a substantially smaller volume of electricity from the local electricity market but covers the largest share of its demand compared to the other consumers.

6.1.1. Minimum maximum indicator

In Fig. 4, we can observe the original MinMax, which measures the ratio between the minimum and maximum imports from the wholesale market. However, this original approach provides limited insight into the situation in the local electricity market, as it yields almost equal values for all cases. The values for the new, separate indicators for prosumers and consumers are also summarised in Fig. 4. The grey fields at the top of the consumer MinMax result from the absence of consumers in the local electricity market for these specific cases.

The new indicators are directly relevant to the values in Fig. 3. Starting with the bottom row of the prosumer MinMax, all the prosumers (PV and battery owners) are able to sell close to 100% of their available surplus within the local electricity market rather than to a retailer. Thus, the MinMax value is 1 or close to 1 for all these cases. For the consumers, the difference in volumes bought from the local electricity market is substantial, resulting in low MinMax values. For example, with one PV and one battery in the local electricity market, House 4 buys an average of 10.5 kWh from the market each month, while House 8 buys an average of 93 kWh. This results in a MinMax value of 0.11 and is a clear indicator of the unfairness in the energy allocation among the consumers in this case.

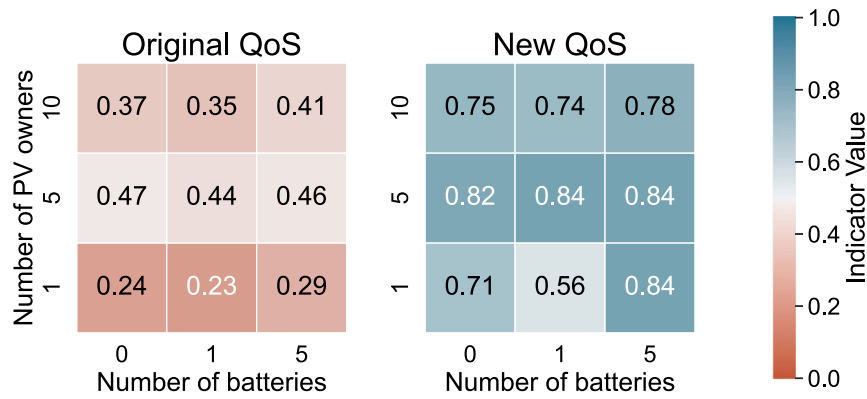


Fig. 5. Monthly average original and new QoS values for the nine case studies. A value of 1 indicates a completely fair market.

It is also evident from the prosumer MinMax that as the prosumer ratio rises in the local electricity market, the local electricity market becomes less fair from the prosumers' point of view. With more surplus and less demand available in the local electricity market, many of the prosumers are forced to sell their surplus to the retailer to a varying extent. This is clear when looking at the shares for these cases in Fig. 3. Although the shares vary with the emergence of batteries, the ratio between the least and most fortunate prosumer is stable, an effect captured by similar MinMax values.

Other relevant redefinitions of these indicators could include measuring the difference in peak import rather than total import. This approach could equip policymakers with critical information about the capacity of local electricity markets to mitigate peak energy consumption, thus also being of value to grid operators. The ability to curb peak energy usage is frequently cited as one of the primary advantages of establishing local electricity markets [57]. This would be particularly relevant for local electricity markets including demand response or other means of flexibility.

6.1.2. Quality of service

In Fig. 5, resulting values for both the original and the proposed QoS are presented. Starting with the original QoS, cases involving only one prosumer are flagged as the least fair. The QoS indicator can be used to pinpoint market participants who have significant influence over outcomes, but in some situations, this identification can be challenging. With only one prosumer present, their high "market power" due to local monopoly is seemingly captured by the low QoS value. With no battery, the PV owner (House 1) sells 100% of their surplus, and allocation among consumers varies between 16.9–72.3 kWh/month. With one battery, the prosumer still sells 100% of their surplus, but consumers cover more demand (10.5–93 kWh/month). However, the disparities in allocated volumes between individual consumers have grown significantly, as captured by the consumer MinMax, without much change in the QoS value. Consequently, while the original QoS can detect a potentially unfair market situation (monopoly), it falls short of capturing the precise nuances of resource allocation in this scenario. The new QoS indicator provides greater clarity in distinguishing between these cases. Furthermore, the case with one PV and five batteries is indicated as significantly fairer compared to the original indicator. Battery owners are now recognised as prosumers, and the remaining consumers receive a nearly equal allocation of electricity. Additional investigation could explore the feasibility of also separately accounting for battery owners in the indicator calculation.

Both QoS versions achieve their highest overall values when there are five prosumers and five consumers, although their indicator values differ significantly. The original low values stem from substantial variations in the volumes allocated to both prosumers and consumers. However, examining the shares reveals that all parties can sell between 62% and 80% of their surplus, which provides a fairer perspective

than the absolute difference of 74–218 kWh. While consumers cover nearly equal portions of their demand, there is a large deviation in the volumes allocated. With the original QoS, four of the market participants (40%) perceive the market as fair, while the new QoS suggests eight participants (80%) perceive it as fair. This corresponds better with the definition in Section 3.

Similar to the prosumer MinMax, the new QoS indicates reduced fairness as the number of prosumers increases from five to ten. However, when considering individual allocations beyond just the minimum and maximum, the QoS suggest a fairer market than the MinMax. Despite the significant difference between the smallest and largest shares, most prosumers can sell a similar proportion of their surplus, thus achieving higher fairness indicator values. Therefore, it is advisable to use both indicators to gain diverse perspectives on distributive justice.

6.2. Cost allocation

The total cost savings over the year of each market participant for each case study is shown in Fig. 6. This is how much each member saves by participating in the local electricity market rather than the wholesale market alone. As with the energy allocation, both shares and volumes are represented in Fig. 6. Volumes (bars) represent total monetary savings, whereas shares (dots) represent the cost savings in percentage.

It is clear from Fig. 6 that all participants benefit from participating in the local electricity market compared to the wholesale market only in all case studies. Particularly, the cases with five prosumers lead to high cost savings for the consumers. It is also clear that introducing batteries to the local electricity market increases the cost savings of both the battery owners and the other market participants.

6.2.1. Quality of experience

With a QoE value of one, all participants in the market encounter identical electricity unit costs across all time steps. Using the supply–demand ratio pricing mechanism, which yields a uniform price, everyone gets the same price when trading in the local electricity market. In such instances, the QoE indicator is more effective in identifying deviations in market coverage compared to direct deviations in electricity prices. Lower QoE values, meaning that the deviation in perceived price is high, indicate that some market participants must interact with the wholesale market in addition to or instead of the local electricity market. As depicted in Fig. 7, the resultant QoE values for this market are consistently high. This is mostly due to the small difference between the upper and lower pricing bound of the local electricity market, resulting in a relatively small difference between the ones that trade within the local electricity market and the ones that do not.

Overall, the cases with five batteries result in the lowest QoE values. Although market participation is higher in these cases, which could theoretically lead to a higher QoE as consumers are less dependent

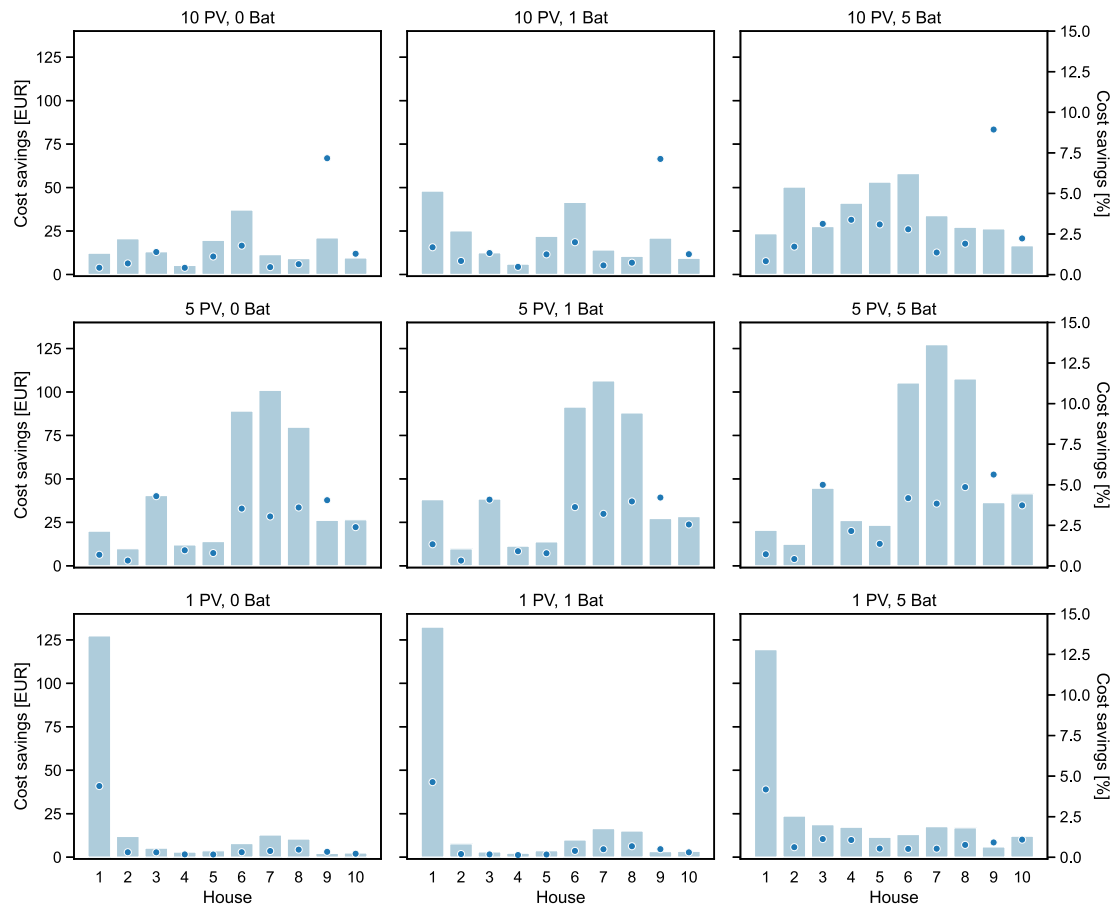


Fig. 6. Annual cost savings of the individual houses per case study when participating in the local electricity market compared to wholesale market only. Bars represent the total amount, and dots represent savings in percentage.

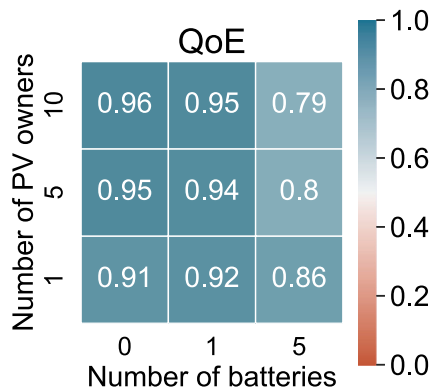


Fig. 7. Annual QoE values for the nine case studies. A value of 1 indicates a completely fair market.

on the wholesale market, the opposite occurs. However, the lower values result from the fact that battery owners are more self-sufficient by using batteries for self-consumption, leading to a perceived cost of zero for that particular time step. As this happens more frequently, the deviation between the participants' perceived prices increases, and the QoE indicator value is reduced.

It is, however, challenging to determine whether the indicator is suitable as a fairness measure overall or if the reason for the suboptimal behaviour is due to the pricing mechanism used in this article. Specific alterations are thus not proposed at this stage. Further investigation should be made using pricing mechanisms with differential prices for

the participants, such as auction mechanisms or distributed optimisation mechanisms. A potential weakness with the QoE is the definition of σ_t^{max} , being the "largest deviation of perceived prices among the participants". With the current assumption, this deviation is always the difference between the wholesale retailer price and the feed-in tariff. In this case study, this difference equals the grid tariff and is thus constant. However, with a dynamic wholesale price or feed-in tariff, this deviation might vary a lot. This can result in the QoE obtaining the same results for very different deviations in perceived prices, which should be taken into consideration when using the indicator as a fairness measure.

6.2.2. Equality index

Fig. 8 shows the resulting values of both the original and the altered version of EI. As with the QoS, the cases that stand out as least fair when using the original EI are the ones with only one prosumer. However, it is questionable whether this is unfair since the prosumer has made an investment that is also beneficial for the consumers. Using the altered version of EI, which considers the equality of cost savings in percentage for the prosumers and in monetary values for the consumers, these cases appear substantially fairer.

The case with ten prosumers and no batteries is indicated as quite fair by the original EI (0.72). However, looking at the cost savings in percentage in Fig. 6 (dots), House 9 seems to have a much higher benefit from participating in the local electricity market than any of the other market participants. This is also the situation in all the other cases with ten prosumers. As seen in Table 3, this is the household with the smallest annual electricity consumption and will thus likely have a relatively large surplus in the hours of solar power generation. By selling this within the local electricity market instead of to the retailer,

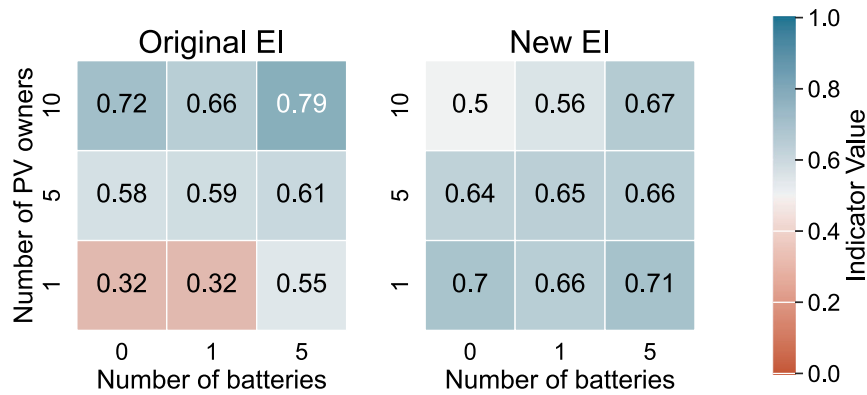


Fig. 8. Annual original and new EI values for the nine case studies. A value of 1 indicates a completely fair market.

the relative cost savings can be substantial.

The new EI also indicates a much fairer market in the case of one PV system and five batteries. This is because households that exclusively own batteries now are being recognised as prosumers by the indicator. The cost savings in percentage these are able to obtain are quite equal, as are the volumes allocated to the remaining consumers. However, it is limited how much value the battery owners add to the local electricity market through arbitrage only. The PV system owner is still the largest contributor in this context and has made a more substantial investment compared to the others. Therefore, it warrants consideration whether segmenting prosumers in the calculation of this indicator would yield a more meaningful result.

Like the QoE, a limitation of the EI index is that it can reach the same value for two very different distributions of cost savings. As it measures the area between the cost saving distribution line and a perfect equality line, it is not able to capture the true shape of the distribution. This should be taken into consideration when using the EI as a fairness measure, both in the original and the adjusted form.

7. Discussion

7.1. Use of fairness indicators in future studies

This study introduces a formalised concept of distributive energy justice for local electricity markets, grounded in the authors' interpretation of the primary aims and incentives for creating such markets. The goals of these markets, however, can differ, leading to a need for different fairness definitions. For instance, consider the following reasons for establishing a local electricity market:

- *Promoting grid services:* If the aim is to promote grid services such as demand response and flexibility, favouring these service providers could be deemed fair. Then, a MinMax indicator measuring the ratio of peak imports could be relevant. More nuanced point-of-view versions of, e.g. the QoS, should be investigated, considering more market participant categories.
- *Encouraging investments in distributed energy resources:* When the main objective is to boost investments in distributed energy resources, a slight bias toward prosumers might be justified. Then, the adjusted indicators proposed in this study should be considered, as they favour equity rather than equality for prosumers.
- *Promoting energy democracy and alleviating energy poverty:* If the ultimate goal is to achieve energy democracy and alleviate energy poverty, an egalitarian benefit distribution emerges as the fairest approach. In this context, the original indicators analysed in this study could still remain relevant as they favour equality among all market participants.

In essence, fairness pursuits should resonate with market participants' motivations, thereby promoting market stability and ensuring their ongoing participation. The definition of distributive energy justice proposed in this article is a combination of the last two points above, i.e., equity and equality. It serves as a starting point for a wider discourse on distributive justice in local electricity markets. It is meant as an encouragement for future research to carefully consider participants' motivations when assessing fairness.

Moreover, the choice of fairness indicators must reflect the market's perception of fairness. If the assessment concludes that absolute equality is the fairest approach, the original indicator definitions from this study remain appropriate. However, as demonstrated here, modifications may be needed to align with prevailing definitions of distributive energy justice. Other indicators not discussed in this study, e.g. those mentioned in Table 1, may also be appropriate for different fairness definitions and should be evaluated in a similar manner as here to confirm their performance.

Fig. 1 illustrates the envisioned practical implications of mature fairness indicators. When the indicators fully align with the definition of distributive fairness and provide appropriate measurements, policy-makers can make regulatory changes to improve fairness. Further, local electricity market operators should use the indicators to adjust their market clearing and pricing mechanisms accordingly.

7.2. Policy implications

Although the resource allocation within the local electricity market itself fulfils the definition of distributive energy justice, the prosumers may be unwilling to participate due to a lack of investment returns. Norway's current feed-in tariff policy is used in the analyses in this article. Prosumers, under this policy, receive a tariff equivalent to the spot price and are exempt from paying the fixed part of the grid tariff for energy supplied to the grid. Consequently, prosumers can generate substantial income from selling surplus energy both within and outside local electricity markets. However, the relative income of prosumers significantly diminishes when the investment costs of PV systems are taken into account. A reduction in the feed-in tariff could thus swiftly transform the financial benefit into a deficit. To uphold the distributive justice of local electricity markets, it is crucial to sustain prosumer investment in production capacity. If feed-in tariffs are decreased, as has been the case in several countries, the market, as conceptualised in this study, may not independently ensure prosumer capital recovery. This issue is also addressed by [11], stating that a lack of attractive feed-in tariffs can restrict both participatory and distributive justice as local communities are unable to benefit from participating in local energy schemes. Instead, alternative external incentives, such as tax deductions or fixed investment subsidies, may need to be implemented to guarantee viability. This approach is currently endorsed by the

Norwegian government through Enova, although the subsidy amount is anticipated to decrease in the forthcoming years due to the maturation of the PV market. It is important to note that this applies to any PV system investment in this context, irrespective of participation in a local electricity market.

The same principle applies to investments in battery capacity. As shown in [58], the payback period of residential batteries can exceed the battery's lifetime. It can thus be assumed that a prosumer with a battery never achieves a positive income by participating in the local electricity market when considering the investment costs. However, it is clear from Fig. 6 that the presence of batteries in the local electricity market enhances the income of all other market participants, including both consumers and prosumers. From a distributive justice perspective, policy-makers should recognise that subsidising investments in batteries and other distributed energy resources benefits not only the investor but society as a whole.

8. Conclusion and outlook

Ensuring energy justice within local electricity markets is not only crucial for sustainable real-life implementation of these markets, but also to align with the United Nations Sustainable Development Goals of affordable, clean energy (Goal 7) and sustainable cities and communities (Goal 11). However, from the literature reviewed in this article, it is evident that there is a lack of explicit definitions of distributive energy justice in a local electricity market context. This article, therefore, proposes a definition of distributional energy justice within the local electricity market. We consider local electricity markets as a means of incentivising distributed energy resources while also promoting energy democracy. Essentially, the definition suggests that individuals who play a role in meeting the local energy demand should be appropriately rewarded without hindering the participation of other market participants who may not be able to contribute at the same level. It is also argued that a distributive energy justice evaluation of local electricity markets requires to be analysed from the point of view of the different market participant categories, as the definitions may vary across these groups.

Additionally, in order to bridge the energy justice and techno-economic research of local electricity markets, it is imperative that fairness indicators that align with the perception of fairness within the market should be developed. In this study, a comprehensive analysis of the indicators MinMax, QoS, QoE and EI have been conducted. It is clear from the analysis that the current definitions of MinMax, QoS and EI do not align with the definition of distributive justice in this article, and are more suitable where strict equality is desired. Thus, the need for redefinitions of these indicators has been underscored, including a point-of-view analysis to separate the perceived fairness of prosumers and consumers. QoE warrants further examination of its performance with other pricing mechanisms before making robust conclusions about its suitability as a fairness indicator.

This work serves as a launchpad for a broader discussion of fairness within local electricity markets. Future studies should engage in interdisciplinary dialogues to comprehensively grasp the societal acceptance criteria for these market structures. This will allow for a more theoretically grounded definition of distributive energy justice in local electricity markets. Furthermore, additional investigation is needed to refine fairness indicators, ensuring their effectiveness across diverse market models. Potential barriers to their implications, such as data privacy considerations, should also be further investigated. More concrete evaluation criteria could also be derived to perform a more generalisable and comprehensive evaluation of multiple fairness indicators. When these fundamental concepts are adequately investigated, one can consider the need to develop entirely new, specific indicators for a local electricity market context.

CRediT authorship contribution statement

Marthe Fogstad Dynge: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Umit Cali:** Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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